Mechanism Is Not Meaning: A Verification Methodology for Cognitive-Architecture Prototypes

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***Abstract*—Building a cognitive-architecture prototype invites a recurring confusion between two claims that resemble each other: that a mechanism is correct, and that it carries meaning. The pressure to overclaim is strongest when a system can be graded on inputs chosen to make it succeed. This paper presents a verification methodology that holds the two apart, demonstrated on Psyche, a dependency-free substrate assembled into a closed cognitive cycle. The methodology combines four practices: a separation of duties between building and certifying, loop-closure driven by system-generated input, adversarial and property-based testing, and pre-registration with anti-circularity gates. Under it, the substrate's properties are proven and the cycle is verified as a mechanism, with no cognitive claim. The methodology is then exercised on three attempts at an externally graded cognitive result: semantic priming, the fan effect, and an interoceptive modulation. Each was halted before any graded number, for a structurally distinct reason. The recurring obstacle, the relationship between the mechanism and an external yardstick rather than the mechanism itself, is the finding. The meaning bar was not cleared; the results concern method and mechanism, not cognition.**

***Index Terms*—verification methodology, cognitive architecture, property-based testing, pre-registration, reproducibility, spreading activation**

I. Introduction

A system built toward cognitive claims is under steady pressure to be described as more than it is. Each working mechanism invites a larger reading, and the least demanding way to support that reading is to grade the system on inputs chosen to make it succeed. The result is a recurring confusion between two statements that resemble each other and are not the same: that a mechanism is correct, and that it carries meaning. Holding them apart is a methodological problem before it is a scientific one, and it is the problem this paper addresses. The contribution is a verification methodology for building cognitive-architecture prototypes without the overclaiming, carried from the substrate upward through one such prototype.

The prototype is Psyche.[1] It is a dependency-free substrate, a graph store and a spreading-activation engine, assembled into a closed cognitive cycle. It was built under a longer-term interest in biologically-motivated, non-LLM cognitive architectures, but that motivation is not what this paper evaluates. Two grades of evidence are kept apart throughout. The substrate's properties are proven: deterministic across separate processes, internally consistent through every change, and free of any read that alters stored state. The cycle above it is verified as a mechanism and nothing more; it is not claimed to perceive, to attend, or to mean. Naming it a cognitive cycle borrows a control structure from an established tradition and asserts nothing about cognition.

The methodology is four practices. A separation of duties keeps the role that writes a mechanism from the role that certifies it. Loop-closure drives each stage with the input the previous stage produced, not input placed by hand. Adversarial and property-based testing asserts invariants designed to break a claim rather than to confirm it. Pre-registration with anti-circularity gates fixes a prediction and its direction before any graded number is computed. None of the four is novel alone; their combination, turned on a developer's claims about a system they are building, is the contribution.

The discipline is tested where the temptation to overclaim is strongest: at the step from a working mechanism to an externally graded cognitive result. Three times the project reached for such a result against data it did not author, attempting semantic priming, the fan effect, and an interoceptive modulation. Three times the anti-circularity gates refused the attempt before any graded number existed, and each time for a structurally distinct reason: one claim could not be separated from the construction of the substrate, one named a generating mechanism the engine does not contain, and one rested on a signal that does not vary on the work the system does. That three independent attempts halted at the same place, the join between the mechanism and an external yardstick rather than the mechanism itself, is the finding. The meaning bar was not cleared, and this paper does not present it as cleared; it reports a result about method and about mechanism. Section II describes the substrate and the closed cycle, Section III the methodology, Sections IV and V the three attempts and what they share, and Sections VI and VII the limitations, related work, and conclusion.

II. The Substrate and Closed Cycle

Psyche is built in two layers carrying two grades of evidence. The lower layer, the substrate, is a graph store with a spreading-activation engine; its properties are proven,

[1] Psyche (developed in the repository under the working name Nous).

in a precise sense: identical output across separate processes under randomized hashing, mutual consistency of the graph indices after every mutation, and reads that never alter stored state. The upper layer assembles the substrate into one loop, perceiving, activating, clustering, selecting, pruning, and writing, whose stages are verified as mechanisms but make no claim to reproduce a cognitive phenomenon. The loop is called a cognitive cycle because its shape is taken from the global-workspace tradition [1], [2], as a structural template; the term names a control structure and asserts nothing about cognition.

The store is a directed graph whose nodes are concepts and whose roads are weighted, directed connections carrying a strength, a usage record, and a decay with elapsed time. The load-bearing decision is that reading and writing are separated at the level of decay. An earlier arrangement applied decay at the moment of access and wrote the decayed value back, with two consequences a casual reading misses: every read mutated the store, and, because the same step re-evaluated closure, a run of reads over time could let a closed road drift back open although nothing crossed it. Psyche instead splits decay into a pure view and a persisting write. A read returns a decayed copy computed from elapsed time but stored nowhere, so reading changes nothing, while only active operations persist a decayed value, and closure is re-evaluated only on an active crossing, never by elapsed time or by a read. The substrate is proven to hold these properties: reads are pure, closure does not drift, the indices stay consistent through every insertion and deletion including on adversarially constructed graphs, and a deletion leaves no dangling entry.

Activation is the substrate's one dynamic operation: given seed nodes, the engine propagates a diminishing quantity along the roads and reports how much each reached node accumulates. The mechanism is the spreading activation of ACT-R [3], borrowed without that architecture's empirical commitments, and two choices make it well defined. The first is bounded single-expansion: each node expands once, at the first hop it is reached, carrying the full sum of all equal-length paths, so activation that converges is summed once and exactly rather than re-counted downstream of the convergence. The second secures determinism: the engine expands hop by hop and freezes each layer before it acts, so no node observes another's expansion within a layer and the outcome cannot depend on the order in which nodes or roads are visited. The engine is in consequence deterministic across separate processes under randomized process hash seeds (5–8 per suite, in fresh interpreters) and, like the store, computes as a pure read. Together the store and the engine form the proven substrate; their two suites contribute 45 and 35 assertions, 80 in all, within a verification suite that stands at 20 suites and 1,372 assertions with no failures.

Above the substrate, Psyche arranges these operations into a single loop (Figure 1). A cycle begins by perceiving:

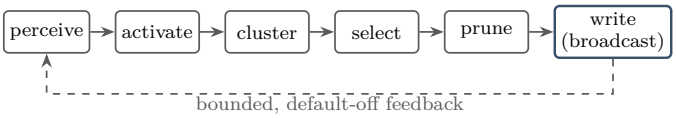


Fig. 1. The closed cycle of Psyche at the level of mechanism: perception and activation, clustering of the perceived sources, selection of one coalition, bridge pruning, and a single bounded write, the broadcast, that biases the next cycle. The feedback is bounded and off by default. The figure depicts graph operations and one bounded write, with the verbs select, prune, and write, not decide, attend, or experience, and makes no cognitive claim.

the input text is matched against the substrate, and the named nodes become the cycle's sources. This is what makes the loop a genuine test rather than a staged one, because the sources are produced by the system from its own contents rather than placed by hand. Because perception names sources that lie close together, the first stage groups them before any field is built, merging by single linkage [4] on road strength alone, so the grouping is topological and does not encode the answer the later stages compute. The engine then builds its field over the grouped sources, the co-activated region is partitioned into candidate coalitions, and exactly one is selected by a competition normalized for coalition size, so a coalition does not prevail merely by being large. Selection then prunes any region the winner reaches across a single bridge into territory the input never named, keeping a reached element only where it attaches at least as strongly to the named region as to the unnamed one. The rule is parameter-free and its one divergence is on the safe side: an unnamed region bridged at least as strongly as it coheres internally presents no valley to cut and is retained, so the rule may over-include an unnamed region but never prunes a member the input named.

Up to here the cycle has only read the substrate. Its one write is the broadcast: the winning coalition is reinforced on its internal roads, biasing the next cycle toward the same region. The reinforcement reuses the store's strengthening rule, which adds to a road's strength a fixed fraction of the distance remaining to its ceiling,

$$s \leftarrow s + \gamma\,(1 - s), \qquad 0 < \gamma \leq 1, \tag{1}$$

so each application closes part of the gap to $s = 1$ without ever reaching it. Boundedness is therefore structural rather than imposed: the broadcast carries no cap, no ceiling test, and no tunable magnitude of its own. With the broadcast off the loop is a pure read whose output is byte-for-byte identical to the pre-broadcast pipeline; enabling it adds exactly one bounded write per cycle. This is the precise content of the claim that the cycle closes: a pipeline that runs to completion on its own output, deterministic and pure in its read portion and bounded in its single write. Each stage, the clustering, the competition and pruning, and the bounded broadcast, has been independently and adversarially re-verified for determinism across random-

ized process hash seeds, integrity, and read-purity, and for none of them is any cognitive claim entered.

III. The Verification Methodology

The substrate and the cycle are the case study; the contribution is the discipline that determined what could be claimed about them, and at what grade of evidence. That discipline counters one hazard: a project building toward cognitive claims is pressed to read more into a working mechanism than it has earned, and the easiest way to do so is to grade the system on inputs chosen to make it succeed. The methodology is four practices that force every claim to earn its tier and hold two statements apart, that a mechanism is correct and that it carries meaning, the first settled against hard properties and the second against external data.

Two qualifications belong at the front. First, the four roles described below are AI subagents operating under enforced lane rules within a single project directed by one person. They are a separation of roles and contexts, not independent human teams and not external auditors. Second, the discipline is reported as what it did, not as a guarantee: it caught the errors recounted below and refused three premature claims, but it is demonstrated on one system and establishes neither freedom from bugs nor that it would transfer.

The work is divided among four roles. An architect specifies each component together with the external criterion by which it will be judged. A builder implements each specification test-first. A separate tester then re-verifies the result, writing its own hostile and loop-closed suites and never editing production code. A validator enters only when a cognitive claim is at stake, grading it against external data under a prediction fixed in advance. The reason for the division is that the role which writes a mechanism should not be the role that certifies it: when author and certifier coincide, verification decays into self-attestation, the tests coming to encode the author's own model of the code and passing for the same reason the code was written. Holding the tester apart makes agreement between builder and tester informative, because the tester was trying to falsify. The tester's checks are largely property-based, asserting invariants that must hold across inputs of a kind, such as determinism across processes, purity of reads, and integrity of the graph, in the tradition of property-based testing [5], [6].

The second practice governs the inputs a test may use. A test that hands a stage the input on which it is expected to work can pass while the stage is structurally wrong, because it never confronts the stage with anything it was not built to handle. Loop-closure requires instead that each stage be driven by the input the previous stage actually produced. This removes the author's choice of input, which is where self-deception enters: as long as the sources feeding a stage are placed by hand, "it works" means only "it works on my guess of its input." Closing the loop forces each stage to consume the previous stage's real output, and it was this substitution that exposed the errors below, several of which were invisible on friendly inputs.

Applied together, adversarial re-verification and loop-closure caught a series of structural errors, defects in the shape of the algorithm rather than the spelling of the code. Six are on record; three are recounted here, and the claim is only that they were caught on this system, not that the discipline finds every error. The first concerns the boundary between reading and writing: computing decay at the moment of access and writing it back made every read a mutation and, because the same step re-evaluated closure, let a closed road drift open over a run of reads. A happy-path test passes this by, since reading the right value is what it checks; the defect was caught by an assertion that a battery of reads leaves stored state identical, and repaired by the pure-view and persisting-write split described above. The second lay not in a mechanism but in how one was judged: an early community-recovery test [7], [8] reported success only because a source had been placed by hand at the centre of each community, so the answer was written into the input. Closing the loop caught it; when sources were instead derived by perception from real input, the same pipeline produced an incoherent result, and the original failure was preserved as a standing assertion set to turn green only once the upstream grouping was genuinely fixed, which it since has. The third was carried from a neighbouring problem: where activation reaches a node along several paths the contributions must add, but a shortest-path intuition re-expands the node and counts everything downstream of the convergence more than once, fixed by bounded single-expansion with a locking check that the post-convergence value is the exact sum. The remaining three were an order-dependence and an identity leak that made results swing across process hash seeds, both removed by hop-layered expansion with frozen per-layer values and a stable tie-break, and a binding degeneracy in which connected components on a graph connected by construction yield a single group, resolved by reframing to a multi-source field.

The fourth practice governs claims about meaning, taken from the use of pre-registration in empirical science [9]. Before any graded number is computed, the prediction and the direction it must take are fixed, and the artifacts of an experiment, its stimulus lists and selection criteria, are content-hashed before any graded outcome exists; stimulus selection is run blind, consuming only the structure of the data and never the quantity against which it will be judged. If the prediction may be adjusted after the data are seen, almost any outcome can be dressed as a confirmation, and the test loses its power to discriminate. In this project the gates are not a formality: on each of three occasions a cognitive claim was put to them, they returned a refusal before any graded number was computed, because the design could not be populated or

graded without circularity.

Taken together, the four practices form one stance toward the system's own claims. None of this proves that Psyche is correct, that the next error would be caught, or that the discipline transfers beyond this system. What it supports is the narrower claim this paper makes: on this system, the discipline caught the structural errors recorded here and refused three premature claims, and it did so by construction rather than by chance.

IV. Three Meaning-Bar Attempts

Three times the project reached for an externally graded cognitive result, and three times the gate declined to grade it. What makes this evidence rather than anecdote is that the three refusals were not one obstacle met three times, but three structurally distinct obstacles: a claim that could not be separated from the construction of the substrate, a claim whose generating mechanism the engine does not contain, and a claim whose external signal does not vary on a realistic workload. A discipline that halts for three different reasons responds to the structure of each problem, not to a single recurring artifact.

Two points of calibration hold across all three. First, each finding concerns what could be established on this substrate with this instrument, not the phenomenon in the world: none of the three says, or licenses, that priming, the fan effect, or interoception is absent in human cognition. Second, in none of the three was a graded result computed. No effect size, no slope, and no direction of effect was produced; the gate fired at selection or at a feasibility check, before the loop was run for score.

The first attempt was semantic priming, with relatedness drawn from a published corpus of human word-association norms [10] rather than from the system. The difficulty is that the corpus encodes association strength as the weight of the edge between two concepts, and Psyche's substrate is built from exactly those weighted edges, so a naive result could appear to prime simply by reading back the edge weights it was constructed from. The pre-registered design therefore required a weight-blind leverage stratum, prime pairs in which the strong and weak associates carry equal edge weight and are matched for target popularity, so that any measured difference would have to come from the loop. The gate was whether that stratum could be populated. A blind selector, consuming only graph structure and the published strengths and computing no activation, found 7 such pairs against a pre-registered floor of 30, and the fallback statistic would have had to exceed its own ceiling of 1 to register an effect. The calibrated finding is narrow: on this corpus, with both the edge-weight and target-popularity paths controlled, priming-through-the-loop cannot be separated from the substrate's own weight map. That is a statement about separability on this corpus, not a claim that priming is absent.

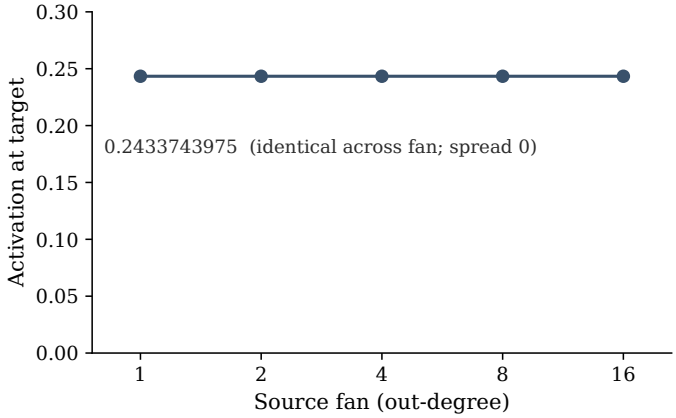


Fig. 2. The fan-effect attempt: the activation delivered to a target is flat across fan size. As the number of a concept's associations grows over the sizes tested, the target's activation does not change to the last digit, because the engine gives each outgoing road the source's full activation and divides nothing by the number of associations. The flatness is a property of the engine as built, not a claim about the fan effect in human memory.

The second attempt was Anderson's fan effect, the finding that the more associations a concept acquires, the more slowly any one is retrieved [11], [12]. It was chosen because it escapes the circularity of priming, riding on the number of a concept's associations rather than the weight of any one. The reason it could not be graded lies deeper, in what produces the effect. In the account it comes from [3], retrieval shares a fixed budget: a source emits a conserved quantity of activation divided among its associations, so each receives less as their number grows. Psyche's engine conserves no such budget; each outgoing road receives the source's full activation, with no division by the number of roads. The consequence was measured: as the number of associations is swept upward, the activation delivered to a target does not fall but stays flat to the last digit, 0.2433743975, across the fan sizes tested (Figure 2). This is a different wall from the first. Priming was circular, and the gate existed to keep a mechanical read-back from being mistaken for a loop effect; here there is no signal to grade in any direction, because the mechanism that would generate one is not in the engine. The two failures are in a sense opposites: the first had a confound to rule out, the second has nothing to measure. An engine change that divides activation among a source's roads would create a fan effect, but it is held as a separate architectural decision, not a repair undertaken to produce the result, and adopting it to obtain the effect would not earn it. The calibrated finding is again narrow: the fan effect is not produced by the engine as Psyche builds it, which says nothing about the fan effect in human memory.

The third attempt left word association behind; its external signal was the machine itself. The cycle was given a body-state scalar, a friction derived from the gap between elapsed wall-clock time and the CPU time consumed, and this friction modulated how far activation was allowed

to spread, in the direction the cue-utilization account associates with load [13] and following a lineage in which bodily state biases cognition [14], [15]. Unlike the first two attempts, the mechanism here is verified: a body-state can change which coalition the cycle selects, and a feasibility pilot confirmed the channel is live, changing the winner on 34 of 40 contrived, flip-capable inputs, a fraction of 0.85. That fraction is a feasibility measure, not a cognitive result. The obstacle is in the signal, not the channel. Friction defined this way rises only when the process waits off the processor, and a realistic Psyche workload does not wait: it is single-threaded, in-memory computation with no input or output to block on, so the CPU time tracks the wall-clock time and the friction sits near zero. Being a ratio, the measure is scale-invariant and does not widen as the graph grows. A range wide enough to grade against could be produced only by manufacturing it, for instance by injecting a pause, which the anti-manufacture rule forbids. This is the third distinct wall: the signal is not circular, as priming's was, and the modulation mechanism is present, unlike the fan effect's, but what is missing is variation. The point is easy to misread as a defect in the mechanism, so it bears stating precisely: the modulation channel is verified and the pilot shows it live; what is flat is the real-world content of the chosen signal. The contention-tracking claim is therefore unsupported on this substrate with this signal, a statement about what this instrument can measure rather than about interoception.

Three attempts, three refusals, and three different reasons: a claim that could not be separated from the construction of the substrate, a claim whose generating mechanism the engine does not contain, and a claim whose external signal does not vary on the work the system does. In each case the gate fired before the loop was run for score, and the response was to decline to produce a number rather than to manufacture one that could not be interpreted. The substrate and the cycle remained verified mechanisms throughout; what failed was the search for an external yardstick they could be graded against without circularity. That the same shape recurred across three unrelated attempts is itself a finding.

V. The Meta-Finding

Read singly, each of the three episodes is a local obstacle. Read together, they show a pattern. In none of the three did the difficulty lie in the mechanism: the substrate computed as specified and the cycle closed as specified throughout. The difficulty lay each time in the relationship between the mechanism and the external yardstick it was to be graded against. Priming failed because the yardstick was the construction itself, the variable to be predicted being the very weights the substrate is made of. The fan effect failed because the engine does not contain the mechanism that generates the effect, so there was nothing for a yardstick to read. Interoception failed because the chosen external signal does not vary on the work the system does. Three different roots, and a single shared locus: not the machine, but the join between the machine and an external measure.

The distinction between a tally and a finding rests on independence. Had the three attempts failed for the same reason, the lesson would be a single nuisance to engineer around. They did not. Circularity, an absent generating mechanism, and a flat external signal are obstacles of different kinds: one is a confound, one is a missing capability, one is a measurement that does not move. That three obstacles so unlike one another arrive at the same locus is what makes the recurrence informative. It locates the genuine difficulty in this style of work. Building a substrate that is correct, deterministic, and closes into a loop proved achievable and was achieved. Finding an external measure that such a substrate can both produce and be graded against, without the measure being a restatement of the substrate, is the harder problem.

The scope of this observation has to be stated plainly. What is established is a pattern on one system, Psyche, across three specific yardsticks. It is not a law about spreading-activation models, about cognitive architectures, or about cognition. Nothing here shows that other substrates face the same join, or that the difficulty is intrinsic to the enterprise rather than to these particular choices. The pattern is real and was observed three times, which is worth recording; it is not general, and is not offered as general. A reader is entitled to take from it a hypothesis about where such difficulty tends to sit, not a result about where it must.

What did and did not hold is worth stating exactly. The proven substrate and the mechanism-verified cycle were untouched by any of the three refusals; their guarantees rest on properties checked independently of any cognitive claim, and no cognitive claim was ever entered against them. What the three episodes record is the discipline operating as designed. The pre-registration and anti-circularity gates exist precisely to stop a claim that cannot be graded without circularity from being graded anyway, and on each occasion they did so before a number was produced. A discipline that halts a result it cannot interpret is not a project failing to obtain one: a refused graded run leaves the tier structure intact, whereas a manufactured one would have corrupted it. Progress on the meaning bar, for a substrate of this kind, depends less on more mechanism than on the construction of external yardsticks that are gradable and non-circular for a system whose content is its own weighted structure.

VI. Limitations and Related Work

Several boundaries bound what this work claims. It was carried out by one person directing a small set of AI subagents, so the separation of duties on which the verification rests is a separation among roles, not among independent investigators; the adversarial roles operate under enforced lane rules and remove the most direct route

to self-confirmation, but they do not constitute outside review. It follows that "verified" carries a specific sense here: an independent adversarial role checked a mechanism against hard properties, determinism across processes, integrity, and read-purity, and found them to hold. It is not a formal proof, and it is not a verification of meaning; a known corner case, such as the one in the pruning rule, is recorded as a standing negative assertion rather than concealed.

The meaning bar was not cleared: across the three attempts no cognitive phenomenon was reproduced, and the positive results are of two kinds only, a methodology result and a mechanism result, neither of them cognitive. The artifact is a prototype, not a system: its core depends on the Python standard library alone, every property was established on small graphs, and no claim is made about behavior at scale. The methodology is demonstrated on a single project, with no control condition and no comparison against development that omitted the discipline, so the evidence is existential. This is an experience report, and claims only that.

Two lineages meet in the work, one behind the methodology and one behind the artifact. The methodology's testing practice belongs to the tradition of property-based testing, in which a test asserts an invariant over a class of inputs rather than a fixed output [5], [6]. The established libraries are more capable; the substrate is dependency-free by design, so the project re-implements a minimal property runner rather than introduce an external dependency, a constraint and not a claim of superiority. The methodology's second borrowing is pre-registration, developed in empirical science to keep a claim from being fitted to its result after the fact [9]; here it governs the grading of cognitive claims about a computational system, and that adaptation is the point of interest.

The artifact draws on a different body of work, and the relationship is one of design lineage, not comparison. The shape of the cycle, a competition among coalitions that yields one winner which is then broadcast, is taken from Global Workspace Theory [1] and its realization as a computational cognitive cycle [2], with the wider global-workspace literature as background [16]–[18]. The spreading-activation mechanism is taken from ACT-R [3]. These supply a structural template and a vocabulary, not a benchmark: Psyche instantiates a control structure of this shape and reproduces none of their cognitive results. The fan effect makes the boundary concrete, since in ACT-R the effect follows from a shared activation budget [11], [12] that the architecture has and Psyche's engine does not; the consequence is a difference in what the two engines contain, recorded as a contrast in mechanism rather than a comparison of performance. The substrate's own mechanisms carry looser analogies to neuroscience, Hebbian strengthening [19] and homeostatic scaling [20], drawn on for design intuition and not as claims of biological fidelity. The interoceptive attempt has a lineage of its own in the idea that bodily state biases what the mind selects [13]–[15]; recent work examines how a machine might carry an interoceptive variable rather than read telemetry [21], and persistent runtimes with ambient self-monitoring are being explored in the language-model setting [22], a different family from the non-LLM substrate studied here.

VII. Conclusion

This paper set out to keep two claims apart, that a mechanism is correct and that it means something, while building toward the second. The means was a discipline rather than a result: separation of duties, loop-closure, adversarial and property-based testing, and pre-registration with anti-circularity gates, carried through the construction of a prototype from the substrate upward. The discipline did concrete work. It caught a sequence of structural errors as the system was built, and it refused three premature cognitive claims before any became a graded number. What stands at the end is a substrate proven against hard properties and a closed cycle verified as a mechanism.

The deepest lesson is about where the difficulty lay. Building a substrate that is correct, deterministic, and closes into a loop proved achievable. The harder problem was the one that remained: finding an external yardstick that such a substrate can both produce and be graded against, without the yardstick turning out to be a restatement of the substrate. The three attempts failed at exactly that join, and differently each time.

The boundary is therefore plain. Psyche is not a mind, and nothing here is a step toward one; no consciousness, no general intelligence, and no validated model of cognition is claimed or implied. The meaning bar was not cleared. What the work offers instead are two results of a narrower kind, one about a method and one about a mechanism, and it has held the line between them and cognition throughout. What remains open is the construction of yardsticks a substrate of this kind can be graded against without circularity, a problem this work locates rather than resolves. The course it commends in the meantime is the one it followed: build what can be verified, claim only what has been, and decline the graded run that cannot yet be interpreted. That is a narrower result than a working cognitive architecture would have been, and a firmer one.

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